χ^2-distribution

normal distribution is a special case of χ^2-distribution when $n=1$.

mgf. χ^2-distribution

$$M_X(t) = E(e^{tX}) = \int_0^\infty e^{tx} \cancel{f(x)}\, dx$$

$$= \frac{1}{2^{n/2}\Gamma(n/2)} \int_0^\infty e^{tx} e^{-x/2} x^{(n/2)-1} dx$$

$$= \frac{1}{2^{n/2}\Gamma(n/2)} \int_0^\infty e^{-\left\{\frac{(1-2t)}{2}x\right\}} x^{(n/2)-1} dx$$

$$= \frac{1}{2^{n/2}\Gamma(n/2)} \times \frac{\Gamma(n/2)}{\left[(1-2t)/2\right]^{n/2}}$$

$$= (1-2t)^{n/2}, \text{ where } |2t| < 1.$$

Charecteristic Function

$$\phi_X(t) = (1-2it)^{-n/2}$$

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Additive property of χ^2 – variate:

The sum of independent chi square variates is also a χ^2-variate. i.e., if $\chi_1, \chi_2, \dots \chi_k$ are K independent χ^2-variate with n_i d.f. respectively then the sum $\sum \chi_i^2$ is also a χ^2- distribution with $\sum n_i$ d.f.

Appointments

Phones to make

* If χ_1^2 and χ_2^2 are two chi square variate n_1 df and n_2 df. respectively then $\frac{\chi_1^2}{\chi_2^2} \sim \beta_2\left(\frac{n_1}{2}, \frac{n_2}{2}\right)$

Student's t distribution. Remarks

Let x_i, $i=1,2,\dots n$ are random samples of size n from a population (normal) with mean μ and variance σ^2.

Then $t = \frac{\bar{x}-\mu}{S/\sqrt{n}}$ & $\bar{x} = \frac{\sum x_i}{n}$

and $S^2 = \frac{1}{n-1}\sum(x_i-\bar{x})^2$ is a